How Long Should an RL Agent Remember?

Continuation and retention regret as a theory of memory use

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Memory and trading off stability and plasticity

Plasticity: ability to stay reactive (forget!!)
Stability: ability to exploit long-lasting regularities (remember!!)
Find right trade-off!
It's all about how to use memory.

Plasticity

Standard tools: Periodic resets, sliding windows, and discounting. Forgetting!

Stability

Forgetting is harmful! Just because it is old does not make information useless.

What does controlling both requirements imply about memory?

Two regrets, one local comparison

After a history $h \in \mathcal{H}$, let

$$R_E^\pi(h,k) = \sup_{\nu \in \Pi^\star} V_E^\nu(h,k) - V_E^\pi(h,k)$$

be regret over the next k rounds.

Continuation regret: plasticity

$$\mathrm{Cont}_{\mathcal{E}}^\pi(T) = \sup_{E \in \mathcal{E}} \sup_{h \in \mathcal{H}} R_E^\pi(h,T).$$

Small regret for T rounds after *every* inherited history.

Retention regret: stability

$$\mathrm{Ret}_E^\pi(n,L) = \mathbb{E}_E^\pi[R_E^\pi(H_n,L)]\,.$$

Small regret after the learner's own length-n history in persistent E.

What do we know about the tradeoff?

Step 1: Choose T (unavoidable!). Step 2: Require continuation near the best possible value

$$B_T^\star = \inf_\pi \mathrm{Cont}_{\mathcal{E}}^\pi(T),$$

Step 3: Subject to this, minimize retention regret.

Let s denote the amount of evidence or testing time used to support a revision.

Evidence obtained while following the optimal action	Persistent retention rate
An observation identifies the environment exactly	0 after a finite transition
Noisy evidence has fixed separation	$e^{-\Theta(s)}$
Noisy evidence has separation $\Theta(1/s)$	$\Theta(1/s)$
A suboptimal diagnostic action is required	$\Theta(1/s)$
Conflicting models are observationally identical	sublinear continuation is impossible

In the active-testing examples,

$$\mathrm{Cont}(T) \lesssim s + \frac{T}{s}.$$

Hence a near-minimax $O(\sqrt{T})$ continuation requirement forces $s \asymp \sqrt{T}$. Tests are initiated at rate $1/s$.

Three observations

1. Plasticity does not justify forgetting. You can do better than forgetting.
2. Plastic agents must have short-term memory.
3. Smart agents must reason about how long to retain memory.

Long- and short-term memory will appear as consequences of balancing stability and plasticity.

Part 1

On forgetting

Forgetting achieves plasticity, but stability asks whether valid information must be discarded.

Forgetting is a serious baseline

These policies all provide age-independent responsiveness:

Periodic restart	Erase the state at block boundaries; restart methods include Rexp3.
Sliding window	Retain only the most recent τ observations; for example, SW-UCB.
Discounting	Give an observation of age u weight γ^u; for example, D-UCB.
Fixed Share	Mix $p_{t+1} = (1-\alpha)\widetilde{p}_{t+1} + \alpha \mathbf{1}/A$.

They are natural first algorithms: simple, analyzable, and used in the nonstationary-learning literature. The question is their retention cost.

Herbster–Warmuth (1998); Garivier–Moulines (2011); Besbes–Gur–Zeevi (2019).

Exogenous forgetting has a retention cost

In E_a, action a always has loss 0; every other action has loss 1.

Complete cold starts
If the next-action law becomes independent of all past evidence at infinitely many ages, then for some a,

$$\mathrm{Ret}_{E_a}^{\pi,\infty}(1) \geq 1 - \frac{1}{A}.$$

Fixed Share, $\alpha = 1/T$
It remains responsive,

$$\mathrm{Cont}^{\mathrm{FS}}(T) \leq \sqrt{\frac{T(1+\log(AT))}{2}},$$

but every action keeps probability at least $1/(AT)$, so

$$\frac{\mathrm{Ret}_{E_a}^{\mathrm{FS},\infty}(L)}{L} \geq \frac{A-1}{AT}.$$

Complete or gradual forgetting discards correct information even when nothing changes.

A retained-state alternative

Full information, A actions, losses $\ell_{t+1} \in [0,1]^A$. Let

$$\Phi(p) = \sum_a \left(p(a) + \frac{1}{A}\right) \log\left(p(a) + \frac{1}{A}\right), \qquad \eta = \sqrt{\frac{\log(A+1)}{T}}.$$

Shifted-entropy mirror descent updates

$$v_{t+1}(a) = \left(p_t(a) + \frac{1}{A}\right) e^{-\eta \ell_{t+1}(a)}, \qquad p_{t+1}(a) = \left[\rho_t v_{t+1}(a) - \frac{1}{A}\right]_+,$$

where $\rho_t > 0$ makes $\sum_a p_{t+1}(a) = 1$.

The policy runs this update continuously. It retains the current decision state and incorporates every new loss vector.

Long-term memory does not impede plasticity

Let $B_A(T) = 2\sqrt{T \log(A+1)}$. For every loss sequence,

$$\mathrm{Cont}^{\mathrm{SEMD}_T}(T) \le B_A(T).$$

This has the minimax order $\sqrt{T \log A}$.

On every constant environment E_a,

$$\mathrm{Ret}_{E_a}^{\mathrm{SEMD}_T,\infty}(L) = 0 \qquad (L \ge 1).$$

Better than forgetting!

Stability favors long-term memory; plasticity does not require forgetting it.

Part 2

Plasticity requires short-term influence

After the same inherited history, different continuations must be able to produce different behavior within T rounds.

Memory is a counterfactual question

Fix the same inherited history h and continue the learner in E_- or E_+. Let

$$P_{E,h}^{\pi,T} = \mathcal{L}_{E,h}^{\pi}(A_m, \dots, A_{m+T-1})$$

be the law of the next T actions, and define

$$\mathrm{Infl}_T^{\pi}(h; E_-, E_+) = d_{\mathrm{TV}}\left(P_{E_-,h}^{\pi,T}, P_{E_+,h}^{\pi,T}\right).$$

Suppose action 1 is optimal in E_-, action 2 is optimal in E_+, and every wrong action costs at least Δ.

The inherited state is identical. Any difference between the two action laws must be caused by observations received after h.

Continuation forces recent evidence to matter

If $\mathrm{Cont}^{\pi}_{\{E_-,E_+\}}(T) \le B$, then after every history h,

$$\mathrm{Infl}^{\pi}_{T}(h; E_-, E_+) \ge 1 - \frac{2B}{\Delta T} = 1 - \frac{2\bar{c}_T}{\Delta}, \qquad \bar{c}_T = \frac{B}{T}.$$

- At fixed gap, smaller normalized continuation regret forces larger T-step influence.
- At fixed regret rate, a larger gap forces a stronger reaction because wrong actions cost more.
- This is general: full-information SEMD already reacts to recent evidence while retaining its state.
- Feedback determines the *cost of obtaining* the evidence, not the need for influence.

Part 3

In search of smart methods

Under partial feedback, a good policy must retain established state and obtain only the evidence still needed to replace it.

Fixed-rate forgetting in bandits

Discounted UCB discounts rewards and counts and selects

$$A_t \in \arg\max_a \left\{ \widehat{\mu}_t^\gamma(a) + 2\sqrt{\frac{\xi \log n_t^\gamma}{N_t^\gamma(a)}} \right\}, \qquad N_t^\gamma(a) = \sum_{u<t} \gamma^{t-1-u} \mathbf{1}\{A_u = a\},$$

where $\widehat{\mu}_t^\gamma$ is the discounted empirical mean and $\tau = (1-\gamma)^{-1}$. Sliding-window UCB retains only the last τ rounds.

On two-armed Bernoulli bandits with gap at least δ,

$$\mathrm{Cont}^{\text{D-UCB}}(T) \le C_\delta \min\left\{ T, \tau \log \tau + \frac{T \log \tau}{\tau} \right\},$$
$$\mathrm{Cont}^{\text{SW-UCB}}(T) \le C_\delta \min\left\{ T, \tau + \frac{T \log \tau}{\tau} + \log^2 \tau \right\}.$$

But on a fixed deterministic bandit, both repeatedly sample the worse arm at rate $\Omega_\delta(\log \tau / \tau)$.

Bounds adapted from the standard D-UCB and SW-UCB analyses of Garivier–Moulines (2011).

Retain an incumbent; test a challenger

Fix a testing scale $s \geq 2$.

1. Keep one incumbent arm. At each incumbent epoch, start a challenge with probability $1/s$.
2. Pull challenger and incumbent in pairs and record

$$D_j = Y_j^{\text{chal}} - Y_j^{\text{inc}} \in \{-1, 0, 1\}.$$

3. Open the test only if $D_1 = 1$. Reject when the cumulative score reaches 0; promote if it remains positive for s pairs.

Every test ends within $2s$ interaction rounds. Established decision state persists; only a bounded block of recent evidence can replace it.

Incumbent–challenger removes logarithmic losses

Measure persistent regret per round by

$$\rho_\nu^\pi = \limsup_{L\to\infty} \frac{\mathrm{Ret}_\nu^{\pi,\infty}(L)}{L}.$$

Tune each policy for adaptation timescale T:

Policy and tuning	Continuation regret	Worst-case persistent rate
D-UCB, $\tau \asymp \sqrt{T}$	$O_\delta(\sqrt{T}\log T)$	$\Theta_\delta(\log T/\sqrt{T})$
SW-UCB, $\tau \asymp \sqrt{T\log T}$	$O_\delta(\sqrt{T\log T})$	$\Theta_\delta(\sqrt{\log T/T})$
Incumbent–challenger, $s \asymp \sqrt{T}$	$O_\delta(\sqrt{T})$	$\Theta_\delta(1/\sqrt{T})$

Localized testing removes logarithmic factors from both continuation and persistent regret.

Mixed feedback environments favor adaptive testing

Before acting, the learner observes an exogenous context

$$X_t \in \{\mathrm{reveal}, \mathrm{bandit}\}.$$

On a revealing round it observes both rewards; otherwise only the selected reward. The reveal process is independent of the current reward pair.

1. Start challenges at rate $1/s$, as before.
2. For at most s rounds, keep playing the incumbent and feed every freely revealed comparison into the sequential test.
3. Only if the test is unresolved, actively collect the missing paired comparisons.

Exogeneity makes free and active comparisons identically distributed.

The smart advantage on benign environments

Uniformly over reveal rates $q \in [0, 1]$,

$$\mathrm{Cont}_{\mathcal{B}_{2,\delta}}^{\pi_s^{\mathrm{EA}}}(T) \le C_\delta \left(s + \frac{T}{s} + T e^{-c_\delta s} \right).$$

If reveal indicators are i.i.d. Bernoulli(q), then

$$\frac{\mathrm{Ret}_\nu^{\pi_s^{\mathrm{EA}},\infty}(L)}{L} \le C_\delta \left(\frac{e^{-c_\delta q s}}{s} + e^{-c_\delta s} \right).$$

Bandit feedback $q = 0$	Fixed reveal rate $q > 0$	Deterministic rewards $q = 1$
$O_\delta(1/s)$ paired-policy rate	exponentially small in s	exactly zero tail rate

The policy preserves the same worst-case continuation order and has a better persistent profile whenever informative evidence arrives passively.

Takeaways

1. **Forgetting is a valid plasticity mechanism, but it is coarse.** Periodic resets, mixing, discounting, and windows incur avoidable retention costs.
2. **Stability favors long-term memory.** Retained state can achieve a minimax-order continuation guarantee.
3. **Plasticity requires short-term influence.** Recent evidence must be able to separate future action laws within T rounds.
4. **Balancing stability and plasticity favors evidence-adaptive testing.** Such policies retain established state and test actively only when passive evidence is insufficient.

Short- and long-term memory emerge from controlling continuation and retention regret together.

Thank you

Questions?



References

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